EAST WEST UNIVERSITY

Department of Computer Science and Engineering

B.Sc. in Computer Science and Engineering Program

Class Test 1, Fall 2025

Course: CSE 207 Data Structures, Section-06 **Date:** October 23, 2025
Instructor: Dr. Nafis Sadeq, Assistant Professor
Full Marks: 10
Time: 20 Minutes

Note: There are **TWO** questions, answer **ALL** of them. Course Outcome (CO), Cognitive Level, and Mark of each question are mentioned at the right margin.

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1. Consider the following C++ code snippet:

```
class Node {
public:
    int value;
    Node* next;
    Node(int v) {
        value = v;
        next = nullptr;
    }
};
```

```
void insertAfter(Node*& head, int key, int value) {
    if (head == nullptr) return;
    Node* curr = head;
    while (curr != nullptr && curr->value != key)
        curr = curr->next;
    if (curr == nullptr) return;
    Node* newItem = new Node(value);
    newItem->next = curr->next;
    curr->next = newItem;
}
```

If the linked list initially contains the nodes `head → [7] → [20] → [5] → NULL` and the function `insertAfter(head, 20, 8)` is executed, **draw the resulting linked list structure and explain how pointer updates occur in each step.**

[CO1, C3, Mark: 4]

2. Given the following sorted array and key:

$$A = \langle 3, 7, 12, 18, 25, 31, 39, 44, 52, 60, 73, 81, 95 \rangle, \quad \text{key} = 44$$

Perform a **binary search trace** showing the values of `low`, `mid`, `high`, `A[mid]`, and the **action taken** at each iteration until the key is found or the search terminates. Finally, state the **index** where the key is found and the **time complexity** of the algorithm.

[CO1, C3, Mark: 6]